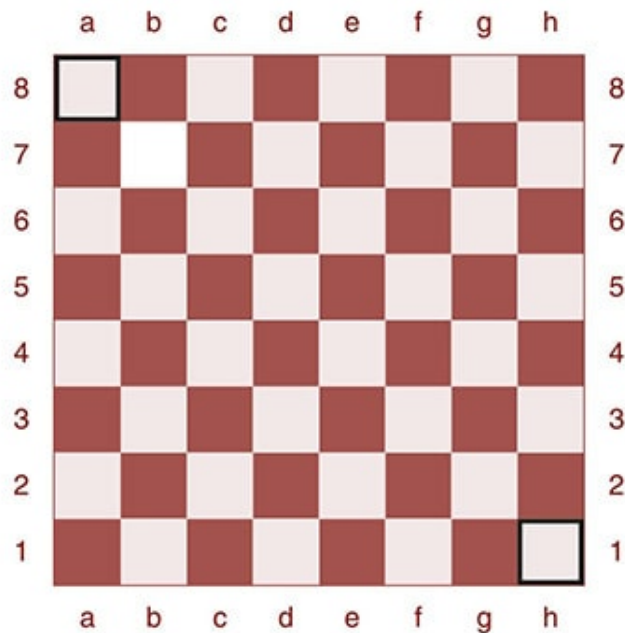


Q-1: Minimum Knight Moves

Anjali and Nakul are good friends. They both had a quarrel recently while playing chess. Nakul wants to know the **minimum number of moves** a knight takes to reach from one square to another square of a chess board (8×8). Nakul is brilliant and he had already written a program to solve the problem. Nakul wants to know whether Anjali can do it. Anjali is very weak in programming. Help her to solve the problem.

A knight can move in the shape of an "L" in a chessboard - two squares either forward, backward, left, or right and then one square to its left or right. A knight move is valid if it moves as mentioned above and it is within the boundary of the chessboard (8×8).



Example Test Cases:

TEST CASE 1

Input:

a1 h8

Output:

6

TEST CASE 2

Input:

a1 c2

Output:

1

TEST CASE 3

Input:

h8 c3

Output:

4

Input Format:

- A function receives two strings:
 - **start_str**: A string representing the starting position of the knight.
 - **end_str**: A string representing the target position of the knight.
- Each string is in the format:
 - First character: A letter from 'a' to 'h' (column).
 - Second character: A digit from '1' to '8' (row).

Output Format:

- The function should return an integer, representing the **minimum number of moves** required for the knight to reach the target position.

Q-2: Everyone Visits Hostel Zero

There are n hostels numbered from 0 to $n - 1$ and $n - 1$ roads such that there is only **one way** to travel between two different hostels (this network forms a tree). Last year, the administration decided to orient the roads in **one direction** because they are too narrow.

Roads are represented by an array `connections`, where `connections[i] = [ai, bi]` represents a road from hostel a_i to hostel b_i .

This year, there will be a big event in the main hostel (hostel 0), and many people want to travel to this hostel.

Your task consists of reorienting some roads such that each hostel can visit hostel 0. Return the **minimum number of roads reoriented**.

It is guaranteed that each hostel can reach hostel 0 after reordering.

Example Test Cases:

TEST CASE 1

Input:

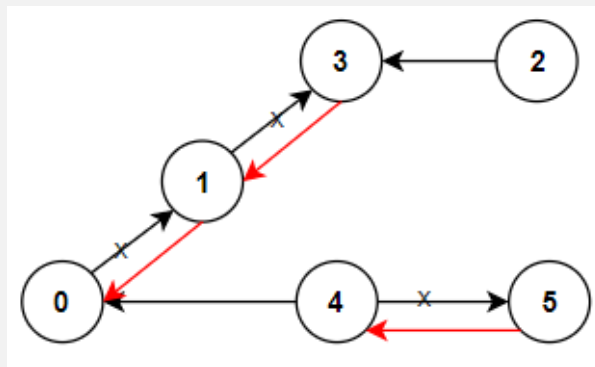
 $n = 6$, $\text{connections} = [[0,1],[1,3],[2,3],[4,0],[4,5]]$

Output:

3

Explanation:

Change the direction of edges shown in red such that each node can reach node 0 (main hostel).

**TEST CASE 2**

Input:

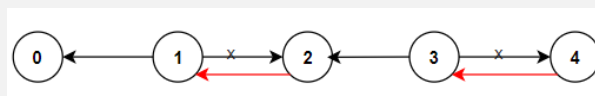
 $n = 5$, $\text{connections} = [[1,0],[1,2],[3,2],[3,4]]$

Output:

2

Explanation:

Change the direction of edges shown in red such that each node can reach node 0 (main hostel).

**TEST CASE 3**

Input:

 $n = 3$, $\text{connections} = [[1,0],[2,0]]$

Output:

0

4

Constraints:

- $2 \leq n \leq 5 \times 10^4$
- `connections.length` = $n - 1$
- `connections[i].length` = 2
- $0 \leq a_i, b_i \leq n - 1$
- $a_i \neq b_i$

Expected Time Complexity: $O(n)$ **Input Format:**

- The function receives the following parameters:
 - **n**: An integer representing the number of hostels.
 - **connections**: A vector of vectors where each element `connections[i]` = `[a_i, b_i]` represents a directed road from hostel a_i to hostel b_i .

Output Format:

- The function should return an integer representing the **minimum number of roads** that need to be reordered so that every hostel can reach hostel 0.

Note: Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.

Academic Integrity Every student is expected to uphold academic integrity and fairness. **Strict action will be taken against any cases of plagiarism or misconduct.**